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Course: Computer Vision

Course code: ITA0506

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Activity - I

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Slot : D

① Medical Image Enhancement

Scenario: An X-ray image shows bones clearly, but the darker tissues have poor quality.

a) Which gray level transformation would you recommend?

Log Transformation

b) Why is it suitable?

- Log transformation expands low-intensity (dark) pixel values.
- It makes details hidden in dark regions more visible.
- At the same time, it compresses higher intensity values.
- At the same time, it makes useful for enhancing darker tissues without excessively increasing already bright regions.

c) What effect will it have on the image?

The dark tissues region becomes brighter and more visible, while the bright bone regions are compressed.

This improves the visibility of details in darker portions of the X-ray.

Log Transformation → Enhances dark regions → makes tissues more visible.

② Night-Time Surveillance

Scenario: Most CCTV image pixel values are in between 0 and 40, resulting in very dark image.

a) Which transformation would improve visibility?

Power-Law (Gamma) Transformation with $\gamma < 1$.

For a dark image, choosing $\boxed{\gamma < 1}$ brightens the image.

b) Why is it more suitable than negative transformation?
Gamma correction increases the brightness of dark pixels while preserving the natural intensity relationship of the scene.

- A negative transformation would reverse the intensities

- Dark $\rightarrow$ Bright

- Bright $\rightarrow$ Dark.

Therefore, the CCTV image would look like a photographic negative rather than a naturally bright night image.

c) Predict the appearance of the enhanced image.

- Parking area becomes brighter.

- Dark objects become easier to identify.

- Cars, people, and other details become more visible.

- The image maintains a more natural appearance.

Gamma Correction ($\gamma < 1$) $\rightarrow$ Brightens dark image $\rightarrow$

Improves night time visibility.

③ Fingerprint Recognition

Scenario: Fingerprint ridge details are present, but

there is low contrast between the ridges and valleys.

a) Which grey-level transformation would you apply?

Contrast stretching

b) Why?

Contrast stretching increases the difference between low and high intensity values.

For the fingerprint:

Before:

Ridges & Valleys $\rightarrow$ Similar Intensities



Low Contrast

After:

Ridges & Valleys $\rightarrow$ Greater Intensity difference



High Contrast

Therefore the fingerprint ridges become more clearly distinguishable from the valleys, helping subsequent recognition.

c) What would happen if a log transformation were applied instead?

Log transformation mainly expands dark intensity values and compresses brighter values. It is therefore more useful for revealing information in dark regions than for uniformly separating fingerprint ridges and valleys.

- It mainly brighten some details, but contrast stretching is more appropriate when the main problem is low contrast.

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